

Moshe Mash, Ph.D.

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Summary

- I am an AI researcher working at the intersection of multi-agent systems, game theory, and human-AI collaboration, with a second, applied line of research in natural language processing and large language models. My research asks how artificial agents and humans make decisions, cooperate, and reach agreements, both on their own and together, combining game-theoretic and mechanism-design analysis with human-subjects experiments and predictive models of human behavior in mixed human-agent settings. I also study how ML practitioners and data scientists build AI systems, capturing and analyzing their workflows to predict where they get stuck. A complementary strand develops LLM-based methods for understanding specialized text corpora, including authorship attribution, citation and cross-reference detection, and textual analysis of legal and patent documents. I hold a Ph.D. in Software and Information Systems Engineering from Ben-Gurion University and was a postdoctoral fellow at Carnegie Mellon University's Robotics Institute. My work received the Best Paper Award at ACM EC 2016 and was selected as a Communications of the ACM Research Highlight (2018), with publications in JACM, AIJ, ACM TIST, CHI, IJCAI, AAMAS, and Science Translational Medicine. I am currently an AI Researcher at DICTA – The Israel Center for Text Analysis and the founder of MashInnovateAI, an independent AI research and development consultancy. Beyond academia, I bring over a decade of hands-on experience taking AI systems from research to deployment, including a predictive system I designed and deployed end to end at Amdocs that remains in operational use, and I have taught at five institutions and mentored students at Ben-Gurion University and Carnegie Mellon.
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Research Interests

Multi-Agent Systems; Game Theory and Mechanism Design; Social Choice and Fair Division; Human-AI Collaboration and Mixed Human-Agent Settings; Decision-Making by Artificial Agents and Humans; Modeling and Prediction of Human Behavior; Analysis of Data Science and Machine Learning Workflows; Natural Language Processing and Large Language Models; Computational Analysis of Specialized Text Corpora (Hebrew and rabbinic texts, legal and patent documents); Human-Centered AI.

Education

Ph.D., Software and Information Systems Engineering | Ben-Gurion University of the Negev | 2014–2018

- **Dissertation:** Reaching Fair Agreements in Group Settings
- **Advisor:** Prof. Ya'akov (Kobi) Gal

M.Sc., Computer Science (AI specialization) | Bar-Ilan University | 2011–2013

- **Thesis:** Joint Exploration with Self-Interested Agents
- **Advisor:** Prof. David Sarne

B.Sc., Software Engineering | Shenkar College of Engineering and Design | 2007–2011

Awards & Recognition

- **Best Paper Award** – ACM Conference on Economics and Computation (2016)
 - **Communications of the ACM Research Highlight** – Which Is the Fairest (Rent Division) of Them All? (2018)
 - **Excellence in Teaching Award** – Ben-Gurion University (2016, 2017)
 - **Ph.D. Scholarship for Academic Excellence** – Ben-Gurion University (2014)
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Academic Positions

Postdoctoral Researcher | Carnegie Mellon University, Robotics Institute | 2019–2022

- Hosted by Prof. Reid Simmons and Dr. Stephanie Rosenthal.
- Conducted research on the behavior and cognitive processes of machine learning practitioners as they develop AI models.
- Designed frameworks to capture, model, and analyze workflows, with emphasis on decision-making, iteration cycles, and knowledge bottlenecks.
- Developed predictive methods to identify critical moments of difficulty (“stuckness”) in ML development, enabling targeted support and improved productivity.
- Collaborated with multidisciplinary teams to integrate human-centric AI approaches into machine learning tools and environments.

- Mentored students in AI and human-computer interaction research.
- Research funded by J.P. Morgan, which renewed the award for a second consecutive year based on the project's strong results.

Research Assistant | Weizmann Institute of Science | 2011-2012

- Developed image-processing algorithms for 3D biomedical images.
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Industry Research Positions

Principal AI Researcher & Founder | MashInnovateAI | 2024-present

- Independent AI research & development consultancy delivering end-to-end custom AI systems - NLP, LLM fine-tuning, RAG, agentic and multi-agent LLM systems, and applied ML - from proof of concept to production deployment for research and industry partners.
- Clients include startups, established companies looking to integrate AI into their products and workflows, and universities and academic research groups.

AI Researcher | DICTA - The Israel Center for Text Analysis | 2022-present

- Lead text-understanding research over large, unstructured Hebrew and rabbinic document collections - turning raw text into structured, searchable, analyzable data.
- Built systems and deep-learning models for in-depth analysis of text - morphological, syntactic, and semantic understanding of long, complex documents at scale.
- Built NLP models for authorship attribution and document classification - predicting the author of texts with historically disputed attribution from textual signals alone.
- Reached high predictive accuracy and applied these models to resolve long-debated questions of authorship, demonstrating reliable classification of ambiguous documents at scale.
- Designed systems for automatic citation detection, extraction, and cross-referencing of sources across a large document corpus.
- Built LLM-based tools for information extraction, semantic search, text similarity, clustering, and question-answering over documents - training custom text embeddings with contrastive learning and fine-tuning LLMs for domain-specific corpora.
- Built end-to-end systems for model training and inference, with tooling to analyze model outcomes and inspect the training process itself.

- Took models from research through to production deployment, delivering document-understanding systems actively used for live analysis of large text collections.

AI Researcher | Diagnostic Robotics | 2019

- Developed machine learning models for the healthcare domain, with applications in risk prediction (e.g., ER diagnoses, hospitalization forecasting).
- Collaborated with clinicians and researchers to refine AI-driven predictive healthcare decision-support systems.

AI Researcher | Amdocs | 2018-2019

- Led the development of an innovative system predicting at-risk maintenance contract renewals, saving the company millions of dollars and still in operational use years later.
 - Designed, built, and deployed the system from scratch, including problem definition, algorithm design, evaluation, and deployment pipeline.
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Teaching & Advising

Ben-Gurion University of the Negev – Software and Information Systems Engineering Dept.

- Teaching Assistant, Databases – Fall-Spring 2016, 2017, 2018
- Teaching Assistant, Analysis and Design of Software Systems – Spring-Summer 2016, 2017, 2018
- Teaching Assistant, Intelligent Systems – Spring-Summer 2015
- Teaching Assistant, Decision Support Systems – Fall-Spring 2015
- Departmental Seminar Coordinator – 2015, 2016, 2017, 2018
- Final Projects Coordinator (4th Year) – 2015, 2016, 2017, 2018
- Final Projects Adviser (Undergraduate) – 2015, 2016

Tel-Aviv Yafo Academic College – Information Systems Dept.

- Lecturer, Databases – Spring-Summer 2015, Fall-Spring 2015
- Lecturer, Game Theory – Spring-Summer 2015
- Teaching Assistant, Databases – Spring-Summer 2014

Bar-Ilan University

- Teaching Assistant, Statistical Methods in Computer Science – Spring–Summer 2014
- Teaching Assistant, Object-Oriented Programming – Fall–Spring 2013, 2014
- Teaching Assistant, Machine Learning – Fall–Spring 2014
- Teaching Assistant, Introduction to Computing – Fall–Spring 2013

Shenkar College of Engineering and Design

- Teaching Assistant, Introduction to Programming Systems – Spring–Summer 2012
- Teaching Assistant, Introduction to Computer Science – Fall–Spring 2012
- Teaching Assistant, Introduction to Object-Oriented Programming – Fall–Spring 2012

College of Management (Academic)

- Teaching Assistant, Object-Oriented Programming – Spring–Summer 2012

High School Teaching (Israel Ministry of Education)

- Mathematics Teacher (2010–2013) – taught Bagrut levels 3–5 points, specializing in students with learning disabilities (via Nitzan Association).
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Professional Service

- **Reviewer:** AAAI, IJCAI, AAMAS, KDD, CHI, EC, ECAI, Group Decision & Negotiation
 - Developed voting algorithm for Best Paper selection (IAAI 2018)
 - Organized departmental seminars at Ben-Gurion University
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Publications

Journal Articles

- Gerhardt D., Marcowitz-Bitton M., Schuster W.M., Elmalech A., Suissa O., Mash M. (2026). Gendered Words and Patent Grant Rates: A Textual Analysis. *Northwestern Journal of Technology and Intellectual Property*, 23(3), Article 2.

- Raphael I, Xiong Z, Sneiderman CT, Raphael RA, Mash M., et al. The T cell receptor landscape of childhood brain tumors. *Science Translational Medicine*. 2025 Mar 19; 17(790): eadp0675.
- Mash, M., Fairstein, R., Bachrach, Y., Gal, K., & Zick, Y. (2020). Human-Computer Coalition Formation in Weighted Voting Games. *ACM Transactions on Intelligent Systems and Technology (TIST)*.
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2017). Which Is the Fairest (Rent Division) of Them All? *Journal of the ACM (JACM)*. *Selected as a Communications of the ACM Research Highlight, 2018*.
- Rochlin, I., Sarne, D., & Mash, M. (2014). Joint Search with Self-Interested Agents and the Failure of Cooperation Enhancers. *Artificial Intelligence Journal (AIJ)*.

Conference Proceedings

- Mash, M., Oryol, S., Rosenthal, S., & Simmons, R. (2022). Predicting Data Scientist Stuckness During the Development of Machine Learning Classifiers. *VL/HCC*.
- Mash, M., Rosenthal, S., & Simmons, R. (2021). DSWorkflow: A Framework for Capturing Data Scientists' Workflows. *CHI*.
- Mash, M., Bachrach, Y., Gal, K., & Zick, Y. (2017). How to Form Winning Coalitions in Mixed Human-Computer Settings. *IJCAI*.
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2016). Which Is the Fairest (Rent Division) of Them All? *EC*. (Best Paper Award)
- Mash, M., Lin, R., & Sarne, D. (2014). Peer-Design Agents for Reliably Evaluating Distribution of Outcomes in Environments Involving People. *AAMAS*.
- Mash, M., Rochlin, I., & Sarne, D. (2012). Join Me with the Weakest Partner, Please. *IAT*.